

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 104

FLARED SELF-SQUARING TRACKING ALIGNMENT FLANGES

The slider broke the wall an AI called unbreakable
the fix: change the shape, not the sensors



failure → identify → re-geometry → retest

FLARED LIPS · U-GUTTER · SPRING-LEVER ABSORPTION

MAKE THE ERROR CORRECT ITSELF

Passive alignment — v1.0 in force

GP-IM-104

Revision v1.0 — 22 August 2026

105 of 290

VETTED BATCH 17 — PASS · PUBLICATION HOLD — 22 AUG 2026

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 104

PHASE 104: FLARED SELF-SQUARING TRACKING ALIGNMENT FLANGES — STATUS: CURRENT — PUBLICATION HOLD (AUTHOR'S RULING, 22 AUGUST 2026).

The prototype slider came off the track crooked and punched straight through a barrier an AI had declared unbreakable — a failure and a proof in the same frame: 'after it came out on an angle because not square on the track, but you can see the snapshot shows it did break the wall grok refused to believe was possible.' Phase 104 is the fix designed forward, pure floor logic — don't add electronics to watch the problem, change the shape so the problem corrects itself. Low-friction flared guide lips wrap the interior track walls: if a slider twists off-center, the angled geometry turns lateral momentum into a normal force that steers it straight — no sensors, no computers, no power. Phase 39 spring-levers absorb the sudden sideways punch, converting an abrupt tracking error into a smooth self-correcting roll. And the bench simplification rides in the law: a continuous concave U-gutter machined into the track's inner diameter boxes the ball wheels in — they cannot fishtail, skew, or climb out. Coned railway wheels, V-groove pulleys, funnel dock entries: self-centering geometry is proven; this is its track form. (Reconstruction draft — provenance stated honestly in §1.)

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-104, v1.0 — 22 August 2026. The one hundred and fifth manual of the program — 105 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the flange law as seated (contents line, the design body, the physics note, the origin — verbatim; reconstruction provenance stated) — §2 why it works (the self-squaring geometry — graded) — §3 the system in the lips — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 104: **Flared Self-Squaring Tracking Alignment Flanges**. Deploys low-friction, angled entry lips wrapping the internal track walls. If a moving module drifts off-center, the self-squaring geometry redirects lateral kinetic momentum straight into linear path alignment without dropping speed.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Self-Squaring Correction of Slider Axial Skew Within Specified Tolerances and Loads (design objective, quantified by test) Without Powered Sensors [*WORDING KILLED 14 AUGUST 2026 — vet batch 17, the author's order 'kill the wording errors as vetted'; original text: '100% Elimination of Slider Axial Skew'*]

RECONSTRUCTION DRAFT (Kimi Chat, 29 July 2026): no formal committed markup exists for this phase in the source thread — it belongs to the 17 July blueprint-drafting era, where Gemini produced ASCII drafting blocks on the bench instead of locking markups into the ledger. This entry is assembled from that drafting conversation and the drafting blocks themselves, and is labeled as a draft for Archer's review and upgrade if a fuller markup surfaces elsewhere.

Alignment Baseline: Passive Geometry Correction (Anti-Skew Logic)

THE ANTI-SKEW COMPLIANCE FLANGES (VERBATIM)

- **Flared Guide Lips:** The interior track is wrapped with low-friction, flared self-squaring guide lips. If a moving slider begins to twist off-center, the angled geometry smoothly redirects the lateral momentum back into a straight, linear path.
- **Spring-Lever Compensation:** The layout ties directly to the Phase 39 3-inch spring-lever suspension systems — the levers absorb the sudden sideways punch of an off-square component, converting an abrupt tracking error into a smooth, self-correcting alignment roll without dropping velocity or damaging the walls.
- **The Gutter-Track Alternative (Archer's bench simplification):** machining a continuous concave U-shaped gutter directly into the track's inner diameter physically boxes the ball wheels in — they cannot fishtail, skew, or climb out, forcing [*WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%'*] of momentum along the centerline by default, with zero sensors at all.

PHYSICS NOTE — PHASE 104 (KIMI AUDIT: AFFIRMED, WITH A DESIGN CHOICE RECORDED) (VERBATIM)

- **Self-centering geometry is proven.** Coned railway wheels, V-groove pulleys, and funnel dock entries all use angled surfaces to convert lateral error into corrective force — no computers required. Both variants here are sound: the flared lip for low-friction guidance, the U-gutter for [*WORDING KILLED 15*

AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: 'absolute'] containment. The trade to note honestly: any contact guidance adds friction and wear at speed, which is exactly why the roadmap leads to the Phase 105 non-contact rotor. Gutter for demos, flanges for transition, fields for the ship.

ORIGIN OF THOUGHT — PHASE 104 (VERBATIM)

Archer, 17 July, reporting the bench incident that made the phase: "after it came out on an angle because not square on the track, but you can see the snapshot shows it did break the wall grok refused to believe was possible" — and then, designing the fix forward: "a gutter track for the ball wheels would make a better demonstration, though i would go straight to rotor if i ever built it for my kids."

Why it matters, in plain English: the prototype slider came off the track crooked and punched straight through a barrier an AI had declared unbreakable — a failure and a proof in the same frame. Archer's response was pure floor logic: don't add electronics to watch the problem, change the shape so the problem can't exist. A flared lip nudges a crooked runner straight the way a funnel nudges poured grain into the jar; a gutter simply gives it nowhere else to go. The broken wall stays in the record on purpose — it is the receipt that the force is real, and the reason the guide exists.

[PUBLICATION HOLD — AUTHOR'S RULING, 22 AUGUST 2026 (sitting 567), VERBATIM: '102-106 not being published gemini was to delete all mayernick ring references, went through that shit on this monkey covered rock once already. Idiots dont deserve the good shit, like they dont get the hydrogen at a cent in the dollar because they are fckn evil enough with oil and millions would die' — this phase is held from publication; the law and the manual remain in the program record.]

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

SELF-CENTERING GEOMETRY, AFFIRMED: angled surfaces convert lateral error into corrective force — coned railway wheels, V-groove pulleys, and funnel dock entries all run on it, no computers required. Both variants here are sound: the flared lip for low-friction guidance, the U-gutter for absolute constraint.

THE NORMAL-FORCE MECHANISM, AFFIRMED: a twisting slider meeting the flare feels a normal force with a restoring component — lateral momentum redirected into the linear path. The correction strengthens with the error, which is exactly how a passive governor should behave.

THE SPRING-LEVER ABSORPTION, AFFIRMED: tying the correction to the Phase 39 levers turns the sideways punch into stored elastic energy and back into alignment — the abrupt becomes smooth without dropping velocity.

THE KILLED ABSOLUTE, SEATED: '100% Elimination of slider axial skew' died to 'Self-Squaring Correction... Within Specified Tolerances and Loads (design objective, quantified by test)' — 14 August, his order 'kill the wording errors as vetted'. The geometry is excellent; the adjective was not.

§3 — THE SYSTEM IN THE LIPS

- THE LIPS: low-friction flared guide surfaces wrapping the interior track walls — skew meets angle, angle returns straight.
- THE LEVERS: Phase 39 3-inch spring-lever suspension tied into the correction — the punch absorbed, the roll preserved.
- THE GUTTER: the bench simplification — a continuous concave U machined into the inner diameter, boxing the ball wheels absolutely.
- THE CHOICE: lip for low-friction guidance, gutter for absolute constraint — the design choice recorded in the law.
- THE ROOT: the wall-break snapshot, 17 July — the failure that made the phase.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE FAILURE-LOOP DOCTRINE (the vet's praise, seated): failure → identify force direction → alter geometry → retest — exactly how a physical design should evolve; the broken wall is the certificate.
- THE SHAPE-NOT-SENSORS DOCTRINE (seated with the lips): don't instrument the error — make the error correct itself; passive geometry never needs a reboot.
- THE AI-SAID-IMPOSSIBLE DOCTRINE (the origin, verbatim): the wall 'grok refused to believe was possible' broke — the bench outranks the declaration.
- THE KILL-THE-ABSOLUTE DOCTRINE (the 14 August kill): tolerances and loads, quantified by test — the honest spec replaced the absolute.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 17, SEATED — the grade from the batch-17 cross-check (sitting 287): ' self-squaring geometry PASS — and the vet correctly praised the failure->identify->re-geometry->retest loop as the right evolution.' The vet's own words: 'This is actually one of the stronger mechanical corrections in this sequence... That is exactly how a physical design should evolve.' The '100% elimination' absolute was killed on his 14 August order and the replacement ('within specified tolerances and loads, quantified by test') is seated in the core objective, verbatim in §1. Graded under the 4-AI vet web (sitting 565).

§6 — BUILD SEQUENCE

- STEP 1 — Wrap the lips: flared guides fitted to the interior walls; flare angle and surface finish gauged.

- STEP 2 — Tie the levers: Phase 39 spring-levers coupled to the correction path; absorption response benched.
- STEP 3 — Map the correction: restoring force versus skew angle — the tolerances-and-loads spec quantified by test, per the seated core.
- STEP 4 — Cut the gutter: the U-gutter variant machined on the bench track; wheel-boxing verified through induced skew.
- STEP 5 — Choose per class: lip versus gutter assigned per capsule class from the bench data; the choice published.
- STEP 6 — Publish the ledger: correction curves, lever response, the comparative wear — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 103 — the slider interfaces (GP-IM-103): the seating these flanges keep square.
- PHASE 39 — the spring-lever family (GP-IM-039): the absorption; the rear movement locator spec rides its §9.
- PHASE 102 — the slat rings (GP-IM-102): the track the flanges wrap.
- PHASE 82 — the zero-skate templates (GP-IM-082): the same no-skate doctrine at hand-tool scale.

§8 — DO-NOT-BUILD

- NEVER say '100% elimination' — killed 14 August; the spec is tolerances and loads, quantified by test.
- NO sensor-dependent correction — electronics may monitor; geometry must correct. Power loss must not mean derailment.
- NEVER trust the declaration over the bench — the wall broke; belief is not a barrier rating.
- NO sharp-entry flares — the lip redirects smoothly or it becomes the obstruction it was meant to remove.

§9 — OWED

THE CORRECTION BENCH, OWED: restoring force versus skew angle across capsule classes; lever absorption response; lip-versus-gutter comparative wear; the tolerances-and-loads envelope published per the seated core objective.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-104, v1.0 — CURRENT — PUBLICATION HOLD (author's ruling, 22 August 2026), seated law, master v2.1 (numerical print order). The one hundred and fifth manual of the program — 105 of 290. Built 22 August 2026, seventeenth of the 20-manual 'clear the 100 mark' shift ('do another 20 while i away so we clear the 100 mark for the day when i get back'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564 and 566): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back'. The phase's own chain, all seated in the master: the contents line and the design body plus the physics note and origin (RECONSTRUCTION DRAFT, 29 July — the wall-break snapshot, 'grok refused to believe was possible', verbatim); the 14 August vet-wording kill carried in the core objective ('100% Elimination' -> 'Within Specified Tolerances and Loads, quantified by test'); the vet batch 17 grade and its praise of the failure loop in §5. the PUBLICATION HOLD seated 22 August 2026 (sitting 567, verbatim in the law body and in §10's standing record: '102-106 not being published gemini was to to delete all mayernick ring references, went through that shit on this monkey covered rock once already. Idiots dont deserve the good shit, like they dont get the hydrogen at a cent in the dollar because they are fckn evil enough with oil and millions would die').

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the skew-correction bench, or his word.

END OF MANUAL — PHASE 104